

NYT Sentiment Analysis Final Report (Team 4)

By: MacKenzie Drury, Yeonwoo Jeon, Selene Nan, Kyle Rozzi, Stefano Poma, Jinseo Yu, and Jane Yijun Sun

Overview

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Project Motivation

- The New York Times, established in 1841, remains one of the most referenced, influential, and trusted publications in the United States
- With nearly two centuries of history, the NYT is widely viewed as the heartbeat of American media, shaping the national consensus
- NYT articles offer a comprehensive, high-quality record of public discourse, making them an ideal source for analyzing how sentiment changes over time

Project Goal

- Develop a reusable, end-to-end framework that:
- 1. Retrieves NYT articles for user-specified keywords and investigation period
- 2. Standardizes and cleans the text
- 3. Computes article-level sentiment scores using VADER
- 4. Performs preliminary data analysis
- 5. Organizes article sentiment into daily time series
- 6. Identifies long-run sentiment trends using simple and volume-weighted moving averages
- This framework is specifically designed to measure how sentiment evolves (for a user-chosen keyword/topic) over time, identifying meaningful shifts in public sentiment

Data Collection

- Our script uses an NYT API key to scan for NYT articles across a specific time range that match with a chosen search keyword, which for this project was ‘Artificial Intelligence Stocks’ and searches and scans up to 10 articles every minute
- The collected data is held in a JSON list of dictionaries with each dictionary entry being a different aspect of the article like the: headline, snippet, abstract, title, and credit

Example:

```
{ "name": "Subject", "value": "live-detached", "rank": 8 } ],  
  "multimedia": { "caption": "An auto factory in China. President Trump has imposed duties on goods from allies like the European Union, Canada and India, as well as rivals like China.",  
    "credit": "Qilai Shen for The New York Times",  
    "default": { "url":  
      "https://static01.nyt.com/images/2025/09/23/multimedia/23biz-global-econ-vlzp/23biz-global-econ-vlzp-articleLarge.jpg", "height": 400, "width": 600 },  
    "thumbnail": { "url":  
      "https://static01.nyt.com/images/2025/09/23/multimedia/23biz-global-econ-vlzp/23biz-global-econ-vlzp-thumbStandard.jpg",  
        "height": 75, "width": 75 } } },
```

Data Cleaning and Preprocessing

After collection, the data needed to be cleaned to make it easier for the sentiment analyzer to use:

- The data was normalized to turn the different dictionary entries into columns
- Duplicate and empty entries were removed
- Text was then lowercase with all punctuation removed to remove unnecessary details that might complicate the sentiment analysis process
- Stopwords were removed because they don't contribute to context and only complicate the sentiment-analysis
- A search keyword is added to the article to record which subject was used to retrieve it
- Then the cleaned data is saved to a CSV file

Data Modeling and Strategies

1. Article-Level Sentiment Modeling

- Combined headline, snippet, and abstract into one text field
- Applied VADER sentiment scoring (best performance on NYT-style text)
- Tested FinBERT for financial topics, but VADER fit NYT writing better
- Each article receives a single compound sentiment score

2. Building the Sentiment Time Series

- Standardized publication dates
- Aggregated article scores into daily and weekly averages
- Used weekly sentiment as the main trend indicator (less noisy)
- Tracked article volume to capture changes in media attention

3. Topic Filtering and Classification

- Used NYT metadata (keywords, section name, type of material)
- Removed irrelevant or loosely related articles
- Ensured sentiment reflects the correct topic (e.g., AI, NVIDIA, politics)

Data Modeling and Strategies

4. Statistical Analysis

- Smoothed sentiment with 7, 14, and 30 day moving averages
- Computed volume weighted moving averages to account for article intensity
- Reduced daily noise to highlight persistent, meaningful shifts in tone

5. Detecting Key Shifts

- Identified weeks with unusually high or low sentiment
- Interpreted jumps in tone alongside events (earnings, politics, tech news)
- Compared sentiment across NYT sections to identify changes in sentiment depending on the type of source
- Compared sentiment trends with external indicators to see whether media tone shifts coincided with financial markets

Case Study: ‘Artificial intelligence stocks’

Keyword Used

- ‘artificial intelligence stocks’

Investigation Period

- April 1st, 2025 to November 30th, 2025

Articles Retrieved

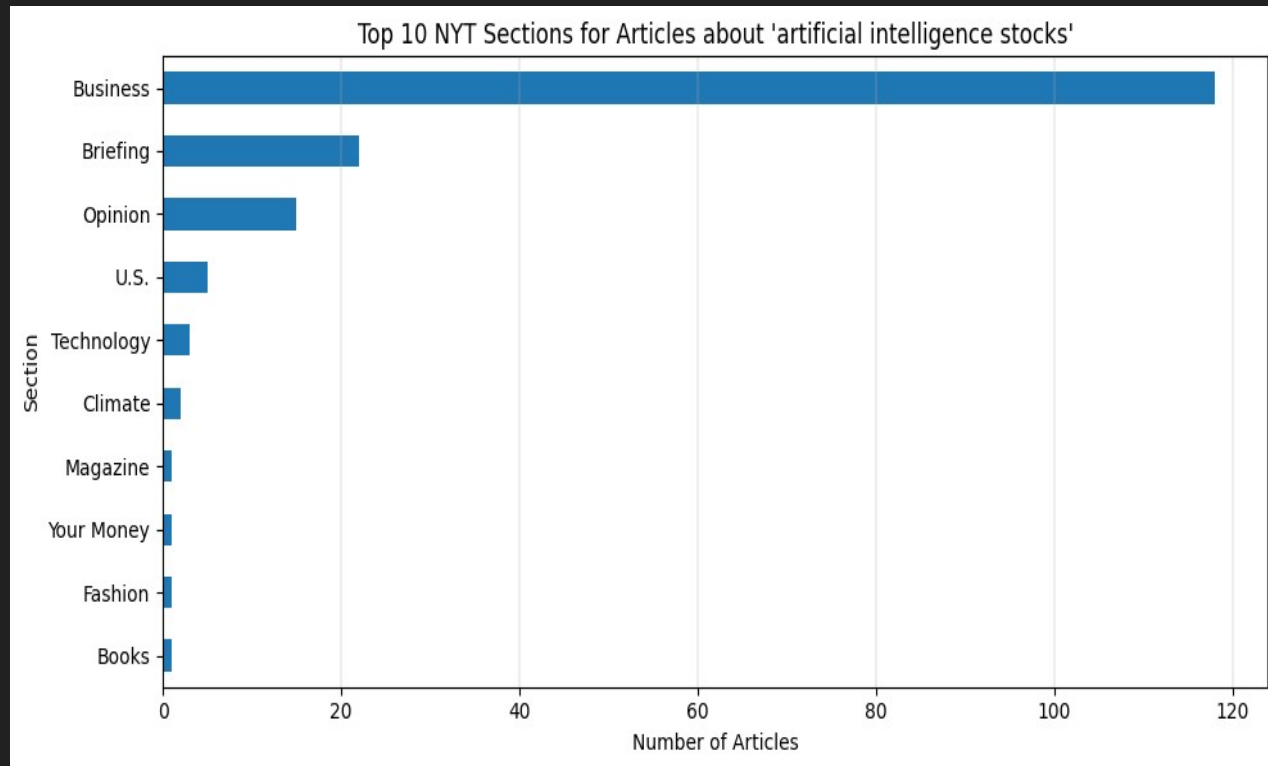
- 171 total NYT articles retrieved

Purpose of Case Study

- Demonstrate the performance of the framework on a real, financially relevant topic
- Evaluate how NYT sentiment evolves relative to AI-sector performance (via AIQ ETF)

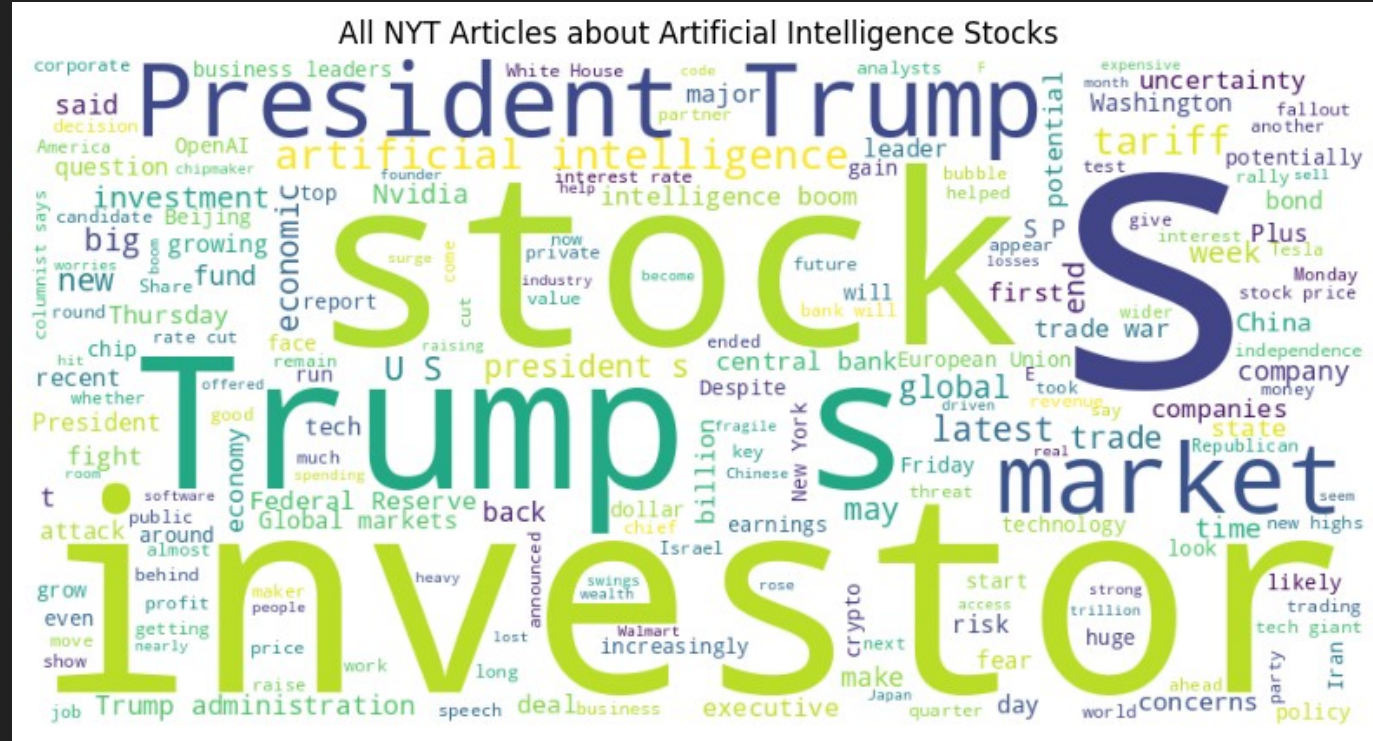
Preliminary Data Analysis - Section Breakdown

Most of the articles containing ‘artificial intelligence stocks’ came from the Business and Briefing sections, showing that NYT primarily frames ‘artificial intelligence stocks’ as an economic or financial story.



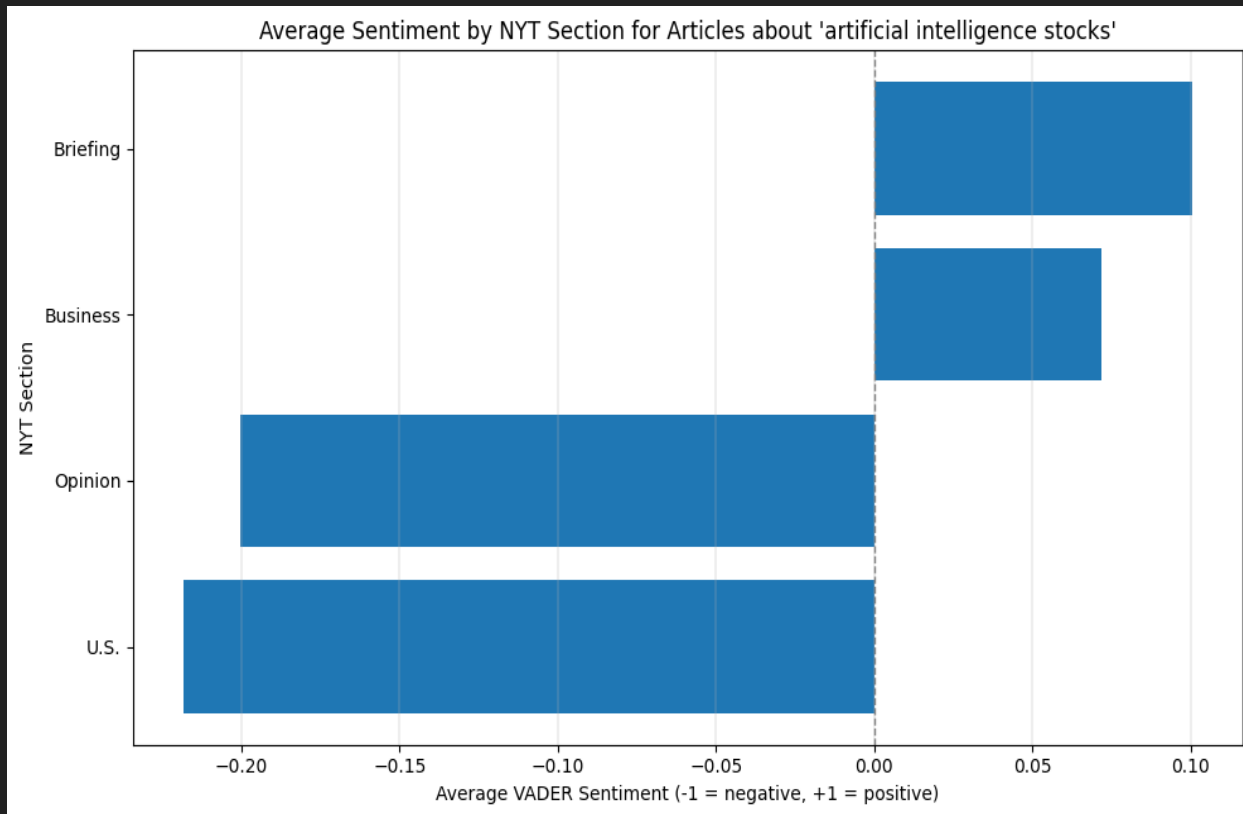
Preliminary Data Analysis - Keyword Patterns

- The keyword patterns show that articles about AI stocks frequently include political and market-related terms such as ‘president’ (72 times), ‘trump’ (68 times) and ‘investors’ (66 times). This suggests that the coverage of AI stocks is often intertwined with macroeconomic and political narratives

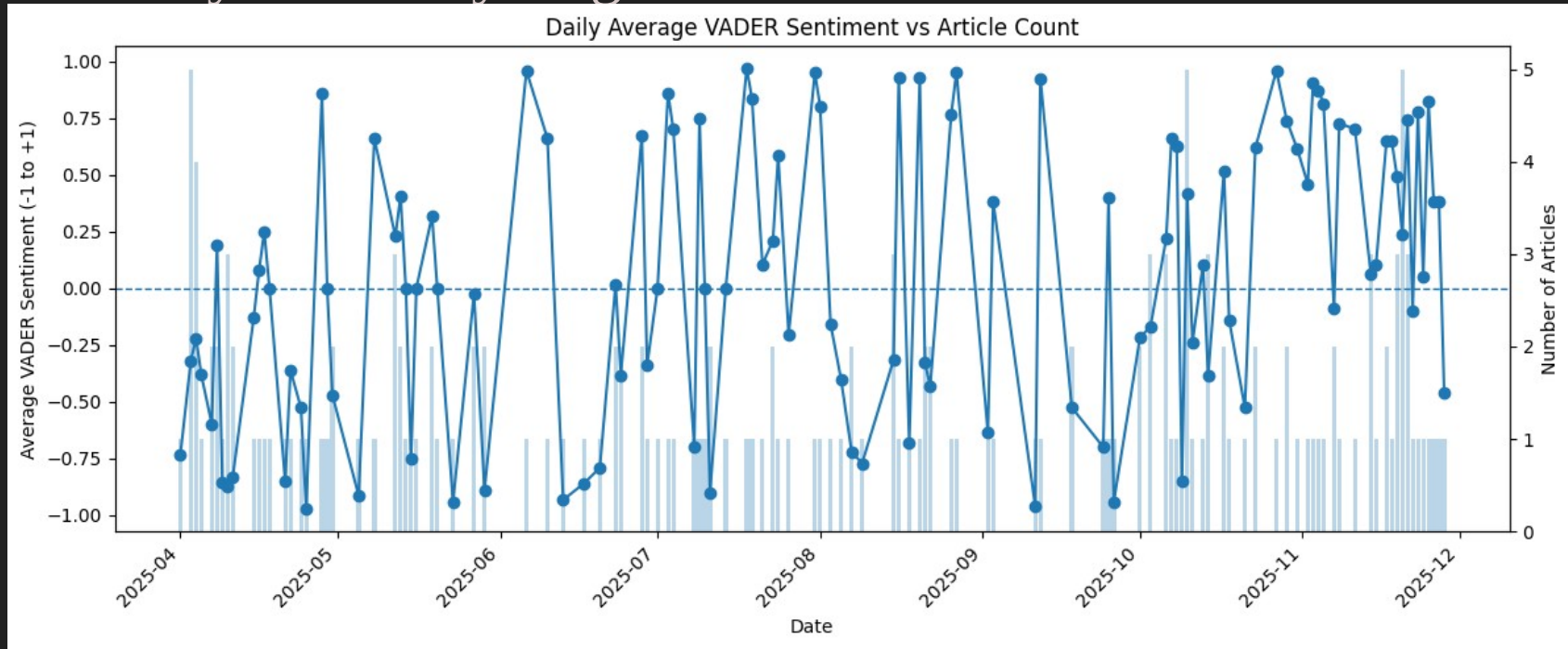


Preliminary Data Analysis - Average Sentiment By Section

Business and Briefing articles tend to have more positive sentiment towards 'artificial intelligence stocks,' while Opinion and U.S. sections lean negative, likely reflecting the contrast between market-focused (economic) reporting and more socially oriented reporting

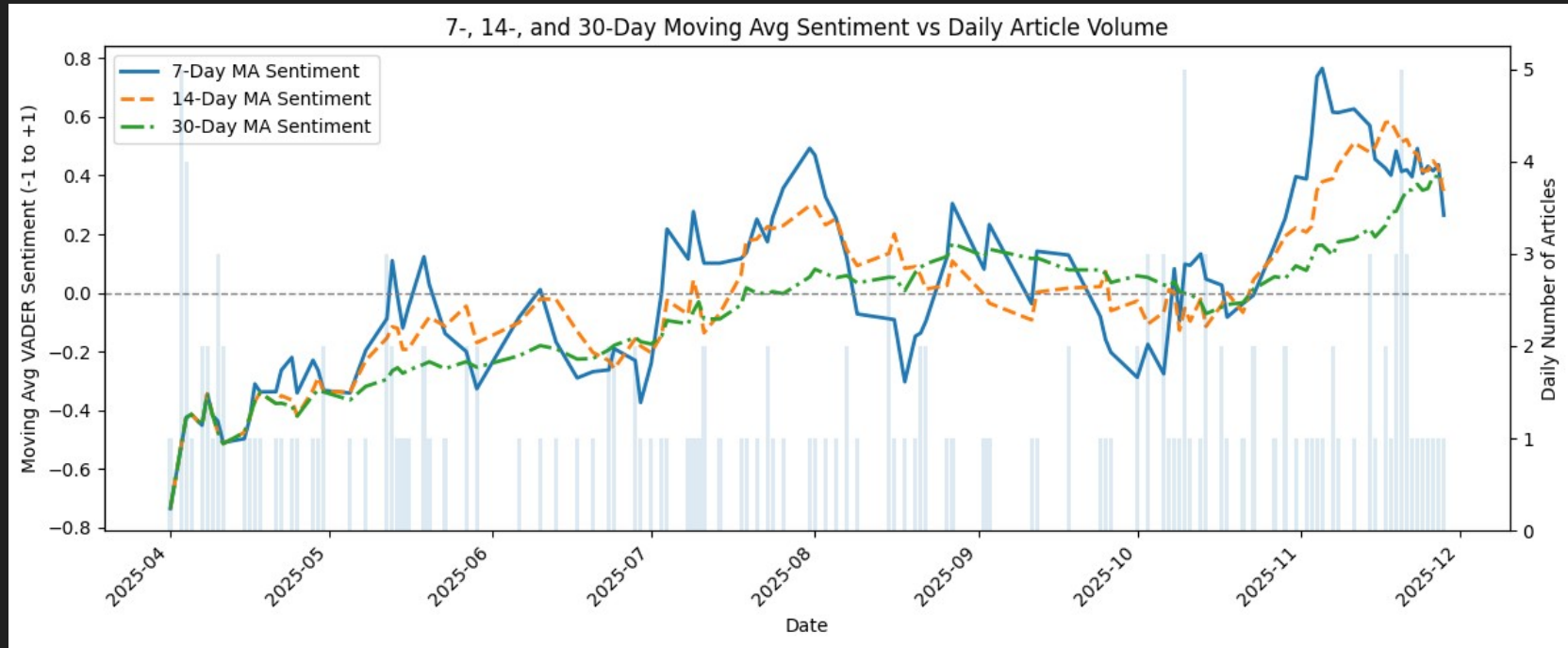


Data Analysis - Daily Avg VADER Sentiment vs Article Count



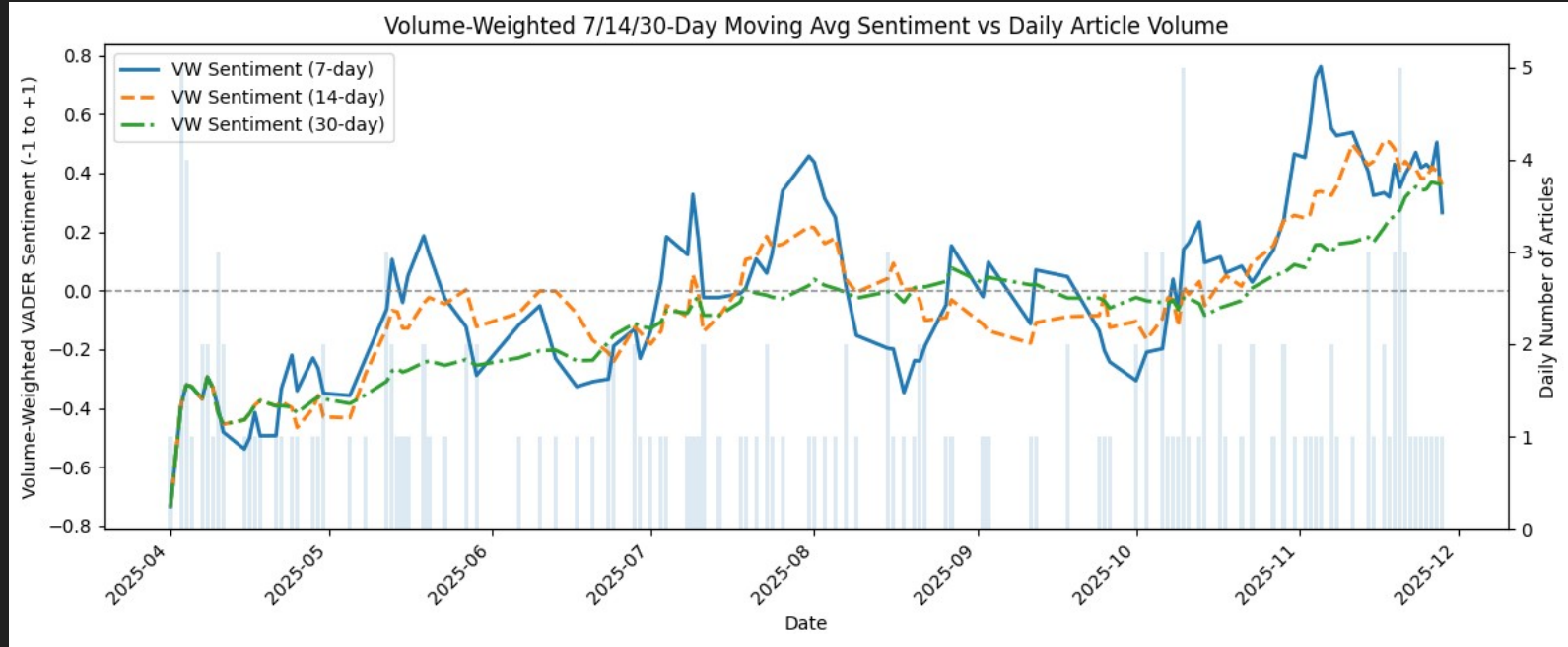
The daily sentiment series is extremely volatile, swinging sharply from positive to negative. This is mainly because most days contain only one or two articles, meaning a single headline can drastically swing the daily sentiment. This instability motivates the use of moving averages to reveal meaningful long-term trends

Data Analysis - 7, 14, and 30 Day Simple Moving Average



Once smoothed via simple moving averages, a clear pattern in the sentiment data emerges: sentiment was consistently negative in the spring (April to June), relatively neutral through the summer months (June to Mid-October), and sharply positive in the fall (Mid-October to the end of November).

Data Analysis - 7, 14, and 30 Day Weighted Moving Average



The weighted moving averages (weighted by article counts) follow nearly the same pattern as the simple averages because daily article counts are generally low (usually 1 to 2 articles per day, never exceeding 5). That being said, the confluence across both methods reinforces our interpretation of a genuine upward sentiment shift across the investigation period.

Comparison to AIQ (AI Stock Index)



AIQ (an index of AI-related stocks) shows a similar overall trajectory to our moving averages over the investigation period: early weakness, stabilization in the summer months, and a strong rally in the fall. Notably, the rally in the fall began around a month earlier in the AIQ than it did in our sentiment charts, indicating that NYT sentiment may lag the financial markets - supporting the idea that improving market conditions may lead to more positive coverage

Conclusion: Key Findings

- Built a reproducible NYT-based pipeline, converting news into sentiment index
- Daily, monthly, and section aggregation showed consistent sentiment patterns
- Business/Briefing sections showed most positive sentiment
- Opinion/U.S. leaned heavily negative
- Moving averages show a clear trend (“Artificial Intelligence Stocks”):
 - Strongly negative (Spring 2025)
 - Neutral recovery (mid 2025)
 - Sharp rally (Late 2025)

Conclusion: Link to Markets and Implications

- AIQ ETF shows similar pattern with early weakness, mid recovery, and late rally.
- Market prices shift **ahead** of sentiment, suggesting that journalism is a **lagging indicator** of markets.
- Media tone reflects and changes with market narratives, rather than predicting them.
- Framework demonstrates that media sentiment can be quantified consistently
- Generalizes beyond AI stocks (this case study is just one example)

Future Work

1. Framework Generalization

Broader Keyword & Topic Coverage

The framework already handles multi-keyword searches, but this could be extended to use embeddings to automatically expand user-defined topics with semantically related terms, synonyms, and ticker-symbols

Multiple Time Periods & Regimes

Apply to pre/post-COVID, 2008 crisis, early AI cycles; check cross-regime sentiment consistency.

Additional Data Sources

Integrate WSJ, FT, Bloomberg, Reuters, Reddit, X/Twitter to capture cross-media divergence.

2. NLP Methodology Enhancements

Improved Sentiment Models

Compare VADER with domain-tuned transformers; validate using manually labeled samples.

Entity- & Aspect-Level Sentiment

Extract sentiment toward companies, regulations, labor, geopolitics → richer multidimensional time series.

Future Work

3. Financial & Time-Series Extensions

Richer Market Variables

Add returns, volatility, trading volume, and option-implied metrics (e.g., sector VIX).

Predictive & Causal Analysis

Use VAR, Granger causality, distributed-lag models to test whether sentiment predicts or reacts to markets.

4. Framework Maturation

Dashboard / Modular Toolkit

Package the pipeline into a reusable toolkit/dashboard enabling user-selected:

- Topics
- Date ranges
- Sentiment models
- Financial comparison series

Transforms the prototype into a general-purpose sentiment analytics framework.